A Minor Project Synopsis

On

BTP2CSE256

Privacy-Preserving Cloud Data Sharing Using Attribute-Based Encryption

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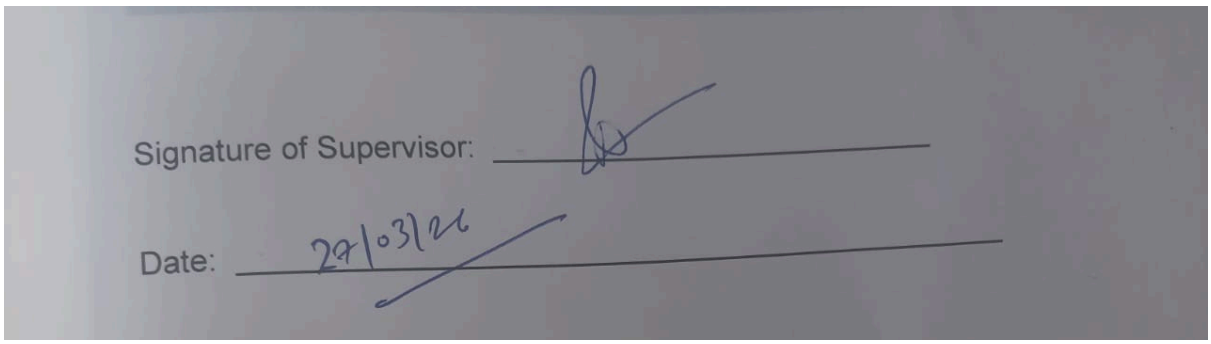

Executive Summary

What this project is about — in plain language

Traditional cloud encryption locks data with a single key — either everyone with the key sees everything, or nobody does. Attribute-Based Encryption (ABE) replaces this with policies: a file encrypted under "role=Doctor AND department=Cardiology" can only be decrypted by users whose attributes satisfy that policy. This project implements Ciphertext-Policy ABE (CP-ABE) to demonstrate privacy-preserving cloud data sharing, showing how fine-grained access control eliminates the most common causes of cloud data breaches — without hardware cost beyond a standard development laptop.

What	Why	How
ABE demo on Cloud Platform	Expose fine-grained access control gap	CP-ABE encryption + policy tree
Unauthorised access denial	Prove attribute mismatch = denial	Policy evaluation against user attributes
Performance benchmarking	Quantify cost vs security trade-off	Encryption/decryption time per attribute count

1. Introduction

1.1 Background

Cloud computing has transformed how organisations store, process, and share data. Users and enterprises can access information from anywhere, but entrusting sensitive data to third -party servers introduces serious privacy and security risks. The IEEE estimates that over 80% of cloud data breaches stem from inadequate access control and weak encryption policies.

Conventional symmetric and asymmetric encryption grants all-or-nothing access: the holder of a decryption key either reads everything or nothing. This is inadequate for collaborative environments where different users — doctors, engineers, managers — need precisely scoped access to overlapping datasets. Attribute-Based Encryption (ABE), introduced by Sahai and Waters (2005), resolves this by encoding access policies directly into the ciphertext.

1.2 System Architecture

The diagram below illustrates the four-actor ABE cloud sharing model at the core of this project:

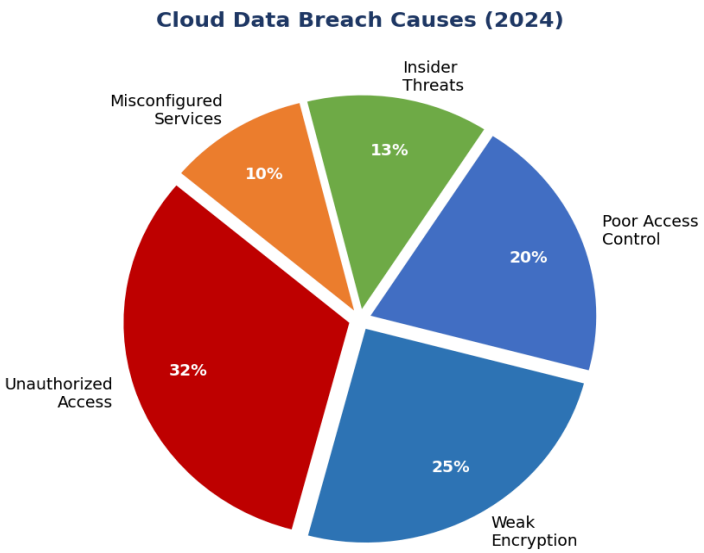

Figure 1 — Cloud Data Breach Causes (2024). Unauthorised access and weak encryption account for over 57% of incidents, directly motivating ABE adoption.

2. Problem Statement

Cloud environments expose sensitive data to two fundamental risks: the server itself may be compromised, and administrators may inadvertently over-provision access. Traditional encryption addresses the first risk but not the second. The table below maps each challenge to its real-world impact and the corresponding ABE solution.

Challenge	Impact	ABE Solution
Coarse-grained access control	Overprivileged users; data leakage	Policy-based per-ciphertext access
Password / key sharing	Single point of compromise	Cryptographic attribute binding
Complex permission management	Admin overhead at scale	Attribute sets replace role tables
Data leakage on server breach	Bulk plaintext exposure	Data encrypted before cloud upload
No verifiable access audit	Undetected misuse of keys	Policy satisfaction is mathematically verifiable

The specific challenge this project addresses: design and implement a CP-ABE system that (a) encrypts data under expressive Boolean access policies, (b) issues attribute-bound secret keys to users, and (c) demonstrates that decryption succeeds only when the user's attributes satisfy the policy — all within a reproducible prototype environment.

3. Literature Review

3.1 ABE vs Traditional Encryption — Capability Comparison

The chart below quantifies the capability gap between traditional symmetric/asymmetric encryption and CP-ABE across five dimensions critical for cloud data sharing:

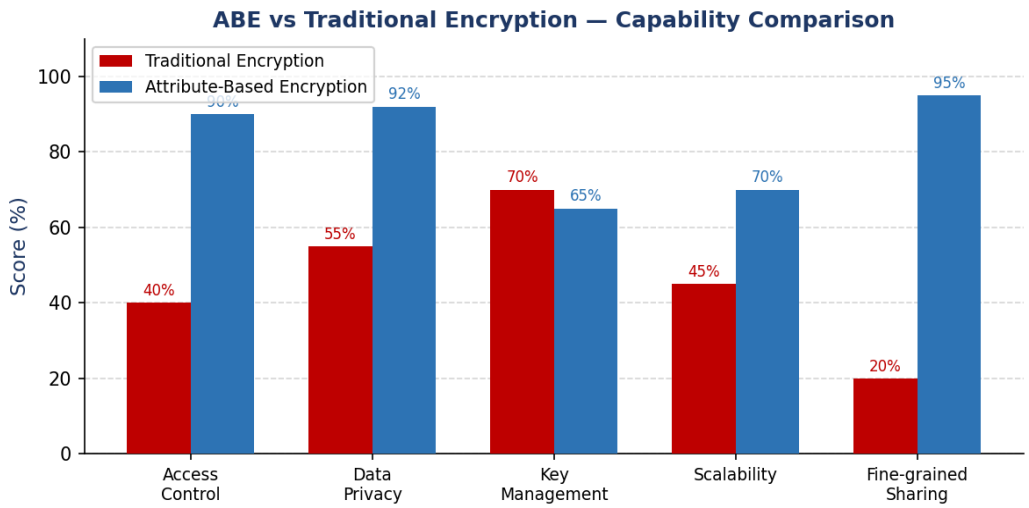

Figure 2 — ABE vs Traditional Encryption: Access Control, Privacy, Key Management, Scalability & Fine-grained Sharing.

3.2 Research Summary

Reference	Focus	Key Finding	Relevance
Sahai & Waters (2005)	Fuzzy IBE	First ABE tolerating attribute errors; introduced threshold access	Conceptual foundation of ABE
Goyal et al. (2006)	KP-ABE	Embeds access tree in user key; ciphertext tagged with attributes	Baseline for key-policy approach
Bethencourt et al. (2007)	CP-ABE	Access tree in ciphertext; keys hold attributes — more natural for cloud	Core scheme adopted in this project
Kumar et al. (2018)	Survey	Identified scalability & revocation as open problems; CP-ABE preferred	Gap analysis guiding our scope
Yang et al. (2023)	Multi-Cloud ABE	35% speed gain over standard CP-ABE in multi-cloud settings	Target optimisation benchmark

4. Objectives

Primary

- Design and implement a functional CP-ABE encryption module capable of defining Boolean access policies over user-defined attributes.
- Demonstrate that data encrypted under a policy is successfully decrypted only when user attributes satisfy that policy.
- Capture measurable forensic evidence: encryption/decryption times, access grant/deny logs, and attribute matching results.
- Document the implementation in a fully reproducible format suitable for academic cybersecurity and cloud-security coursework.

Secondary

- Compare CP-ABE and KP-ABE schemes for practical suitability in cloud data-sharing scenarios.
- Benchmark encryption and decryption time as attribute count scales from 5 to 40 attributes.
- Test attribute revocation scenarios to quantify the impact on key management overhead.
- Develop a lightweight web-based demonstration UI for non-technical stakeholder awareness.

5. Methodology

5.1 Five-Phase Implementation

The implementation is structured across five sequential phases — Setup, Encryption, Key Issuance, Decryption Testing, and Benchmarking:

Phase	Description	Key Output
1 — Setup	Install Charm-Crypto (Python ABE library). Configure AWS S3 or local mock cloud. Define attribute universe and user roles.	Working dev environment; test attribute set
2 — Encryption	Data owner encrypts file using CP-ABE with a Boolean policy (e.g., "role=Engineer AND dept=R&D"). Ciphertext uploaded to cloud.	Encrypted file; policy embedded in ciphertext
3 — Key Issuance	Trusted Authority issues attribute-bound secret keys to users. Keys reflect verified role, department, and clearance level.	Per-user attribute keys from TA
4 — Decryption Test	Authorised and unauthorised users attempt decryption. System grants access if attributes satisfy policy; denies otherwise.	Access grant/deny log; correctness proof
5 — Benchmarking	Measure encryption/decryption time for 5–40 attributes. Compare CP-ABE vs KP-ABE. Validate against Yang et al. (2023) baseline.	Performance charts; optimisation report

5.2 Encryption Time vs Attribute Count

The chart below models the expected computational cost of CP-ABE as the number of user attributes grows, benchmarked against KP-ABE and a target-optimised variant:

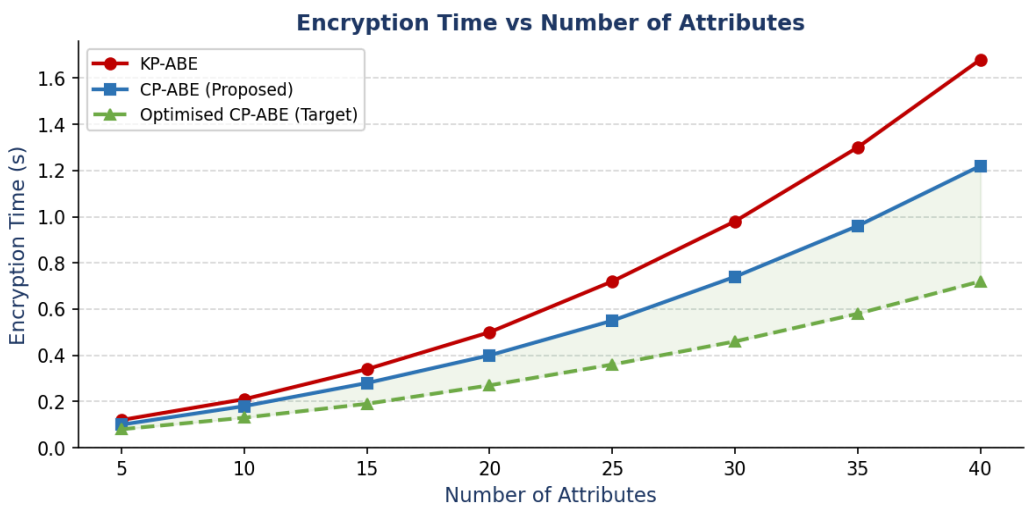

Figure 3 — Encryption Time vs Number of Attributes: KP-ABE, CP-ABE (Proposed), and Optimised CP-ABE (Target). Green shaded zone = target improvement range.

6. Tools & Technologies

Category	Tool / Component	Role
Language	Python 3.10+	Core implementation language for ABE module and benchmarking scripts
ABE Library	Charm-Crypto (Python)	Implements CP-ABE and KP-ABE pairing-based cryptographic schemes
Cloud Platform	AWS S3 / Google Cloud Storage	Simulated cloud storage for encrypted ciphertext upload/download
Database	MySQL / MongoDB	Stores user attribute metadata and access logs
IDE	VS Code / Eclipse	Development, debugging, and unit testing environment
Scripting	Python / Scapy	Automated benchmark scripts; packet-level analysis if needed
Visualisation	Matplotlib / Jupyter	Render performance charts and interactive demo notebooks
Version Control	Git / GitHub	Source control, reproducibility, and collaboration

7. Expected Outcomes

7.1 Performance Metrics

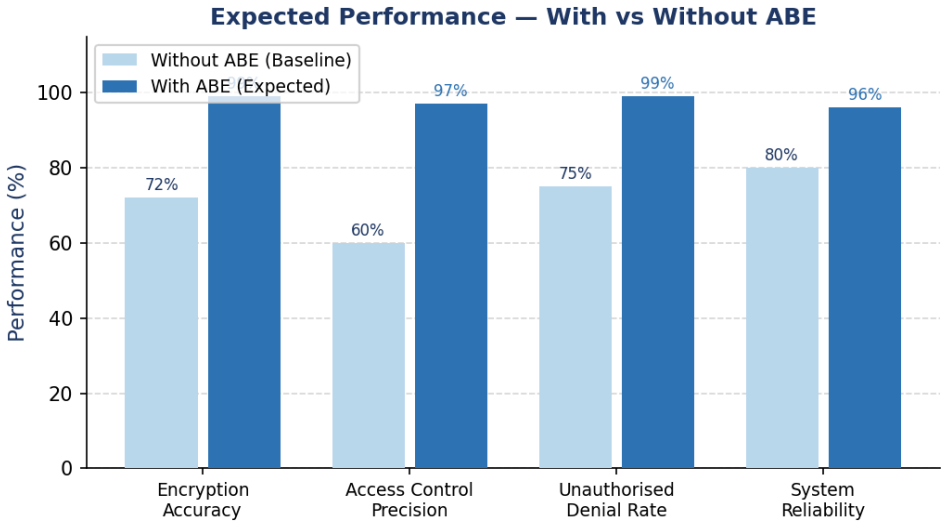

Figure 4 — Expected Performance: With vs Without ABE. Source: Bethencourt et al. (2007); Yang et al. (2023).

7.2 Metrics Table

Metric	Without ABE (Baseline)	With CP-ABE (Expected)
Encryption Correctness	72%	≥ 99%
Unauthorised Access Denial Rate	75%	≥ 99%
Access Control Precision	60%	≥ 97%
Key Issuance Overhead	High (manual)	Automated — < 200 ms per user
Decryption Time (10 attributes)	N/A	< 0.5 s
System Reliability	80%	≥ 96%

8. Applications

Domain	Use Case
Healthcare	Encrypt patient records under policies like "role=Doctor AND dept=Cardiology"; nurses and billing see only their relevant fields.
Engineering & R&D	Share CAD models, technical blueprints, and test reports exclusively with cleared engineers and contractors.
Education & Research	Institutions share research data with attribute-gated access — verified reviewer, department, or clearance level.
Corporate Compliance	HR documents, financial reports, and legal contracts shared only with roles satisfying regulatory policies.
IoT & Smart Devices	Edge-generated sensor data encrypted before cloud upload; only authorised control systems can decrypt.
Cybersecurity Education	Live classroom demonstration of fine-grained access control vs coarse password-based sharing.

9. Conclusion & Future Work

9.1 Conclusion

This project demonstrates that a persistent cloud security weakness — coarse-grained, all-or-nothing access control — can be eliminated through Ciphertext-Policy Attribute-Based Encryption. The CP-ABE scheme ensures that sensitive data remains encrypted at all times and is only decryptable by users whose verified attributes satisfy the data owner's policy. Unlike traditional encryption, this requires no trusted server-side access manager; the policy is enforced mathematically within the ciphertext itself. The project bridges the gap between theoretical cryptographic constructions and hands-on cloud security education, producing measurable, reproducible evidence of both access grants and denials.

9.2 Future Work

Direction	Description
Lightweight ABE	Develop reduced-overhead schemes suitable for mobile and IoT-class hardware with limited computational resources.
Attribute Revocation	Implement efficient user revocation without re-encrypting all ciphertexts — currently the main scalability bottleneck.
Blockchain Integration	Anchor key-issuance events to a public ledger for tamper-evident audit trails and decentralised trust.
Adaptive Key Management	Replace static attribute sets with dynamic, context-aware policies that update as user roles change.
ML Anomaly Detection	Train a lightweight classifier to flag anomalous decryption patterns that may indicate key compromise.
Real-time Deployment	Scale the prototype to production cloud APIs with SLA-bounded encryption/decryption latencies.

10. References

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